

Supplementary material: Masking cannot remove what is inside the region of interest

S1. Pre-registration index and registry

Every experiment was specified in a dated document committed to the public repository before any model was fitted on the corresponding data; amendments were committed before the amended analysis. The registry lists, for every registration, its first commit and the first commit that added outcome results (outcome-free cell counts excluded; for registrations that add arms to an existing results folder, the first commit whose results contain the new arm). Commit times are set by the committing machine and are therefore not an external time stamp; the registrations of the second review round (`PREREGISTRATION_REVIEW2`) were additionally pushed to GitHub before any model was fitted. The primary family used for multiplicity control (`STATISTICAL_PLAN`) was assembled after the individual results were available; each member test was individually pre-registered, and the Holm correction is applied to the whole family. Generated by `scripts/verify/registry_table.py`.

Registration	Registered (commit, UTC)	First outcome result (commit, UTC)	Order
PREREGISTRATION_ISIC2019_TRAPS	76e25d4 2026-09-26 13:14	2699d51 2026-09-26 15:47	registration first
PRECOMMIT_MATCHED_TRAPS	5ef3228 2026-09-26 14:57	2699d51 2026-09-26 15:47	registration first
PREREGISTRATION_CXR_DEVICE_TRAPS	4d1a924 2026-09-26 18:07	e0d5310 2026-09-26 18:44	registration first
PREREGISTRATION_UNIVERSAL_I2E	ee95f10 2026-09-26 20:14	2f09c07 2026-09-26 21:07	registration first
PREREGISTRATION_THYROID_TRAPS	ea57661 2026-09-26 22:46	b7bc5e7 2026-09-27 00:07	registration first
PREREGISTRATION_SLAS	2d1ba20 2026-09-26 23:33	2d1ba20 2026-09-26 23:33	same commit
PREREGISTRATION_CAPSULE_TRAPS	fcbf8d9 2026-09-27 00:40	6c3e8c5 2026-09-27 00:59	registration first
PREREGISTRATION_CXR_DRAIN	3d0cffc 2026-09-27 03:29	1ef04b7 2026-09-27 04:16	registration first
PREREGISTRATION_FINETUNE	7c8b466 2026-09-27 07:06	1f4f6e3 2026-09-27 08:00	registration first
PREREGISTRATION_OVARY_TRAPS	2517600 2026-09-27 07:29	cd67b8e 2026-09-27 07:45	registration first
STATISTICAL_PLAN	4e9d44c 2026-09-27 09:37	4e9d44c 2026-09-27 09:37	same commit
PREREGISTRATION_NATURAL	74bd243 2026-09-27 09:39	be95223 2026-09-27 09:50	registration first
PREREGISTRATION_INPAINT_LAMA	715b790 2026-09-27 09:41	ae374fd 2026-09-27 10:14	registration first
PREREGISTRATION_TEXT_PROMPT	7877e8c 2026-09-27 12:21	2203aa2 2026-09-27 12:54	registration first
PREREGISTRATION_THEORY_PREDICTION	4f78653 2026-09-27 13:46	30ce70a 2026-09-27 14:03	registration first
PREREGISTRATION_EMBEDDING_GROUPS	b0eda23 2026-09-27 13:55	30ce70a 2026-09-27 14:03	registration first
PREREGISTRATION_SCALE	13303b0 2026-09-27 14:13	cef4871 2026-09-27 14:49	registration first
PREREGISTRATION_UMTE_ABLATION	30f19c8 2026-09-27 14:15	0abd4a7 2026-09-27 14:22	registration first
PREREGISTRATION_FT_ERASE	8ac7a7d 2026-09-27 14:20	3d0dd70 2026-09-27 14:36	registration first
PREREGISTRATION_FT_UMTE	b7d7077 2026-09-27 14:52	f28ab78 2026-09-27 15:27	registration first
PREREGISTRATION_REVIEW2	29cd0da 2026-09-27 18:56	3673625 2026-09-27 19:40	registration first
PREREGISTRATION_REVIEW3	85da735 2026-09-27 23:43	e4ffc8a 2026-09-28 00:11	registration first
PREREGISTRATION_ROUND4	d34b8b8 2026-09-28 11:28	ac90660 2026-09-28 12:48	registration first
PREREGISTRATION_ROUND5	4af106a 2026-09-28 13:37	a02db23 2026-09-28 14:01	registration first
PREREGISTRATION_ROUND6	607900c 2026-09-28 14:27	4c3263d 2026-09-28 16:29	registration first
PREREGISTRATION_ROUND7	cbf4229 2026-09-28 20:37	68e2432 2026-09-28 20:49	registration first

Registration status and multiplicity of every analysis family.

Analysis	Registration status	Multiplicity
Real traps, controlled sweeps, natural data, U-MtE (P1–P4)	before results	16-test primary family (Holm)
Primary-family grouping	after results (post hoc)	—
Rebuild on new hardware (R0)	before results	tolerance criterion
Covariate-matched traps (R1)	designed after the real-trap results; registered before its own results	own family (Holm, 4)
Real-artifact transplant (R2)	designed after the real-trap results; registered before its own results	own family (Holm, 4)
Operating points (R3); conflicting subgroup	test registered before computing; subgroup defined post hoc	decision rule
Fine-tuned hair, ovary power (R4)	before results	single tests
Paste-edge control (R5)	designed after the transplant results; registered before its own results	own family (Holm, 4)
Crossed bootstrap (R6), stricter matching (R7)	designed after the results; registered before computing	re-estimation / sensitivity

S2. Replication of the original pilot

Of 88 numbers reported in the thesis proposal, 73 are reproduced exactly (same sign, same confidence-interval verdict, $|\Delta| \leq 0.02$), 12 agree in sign and significance with a larger difference, and 3 differ (a borderline occlusion interval, a contrast trap that was reported as null and is not, and one source stratum). Details: `results/SUMMARY.md`. The archived pilot confidence intervals are recomputed exactly from the archived predictions (`make verify`).

S3. Datasets considered and not used

CANDID-PTX (not accessible outside Health New Zealand); RANZCR-CLiP devices linked to NIH (masking helped at both locations; device placement confounded with other devices); chest-radiograph cardiomegaly with lines over the heart (210 labelled images); TCGA/GrandQC pathology (pen-mark masks unreliable); ISIC ink (70 in-lesion images); breast ultrasound BUSI/BUS-BRA (caliper-free group only $\approx 70\%$ pure after a two-detector design); DDTI thyroid (TI-RADS instead of pathology; 134 images); BKAI-IGH colonoscopy specular highlights (no highlight-free images). *External validation with patient identifiers*. TN-SCUI 2020 (one image per patient, nodule masks) may not be used outside the challenge under its terms. ThyUS2Path [4] (842 patients, 8,508 images, surgical pathology, CC BY 4.0) contains real calipers in several colours but no nodule masks, and our frozen caliper detector responds to elements of its scanner interface (frame lines, depth scales) on a random 25% sample inspected before any label was read. Using it requires a nodule segmentation transferred from TN3K and a caliper detector re-validated on this interface; we did not run it.

S4. Artifact labels: provenance, audits, sensitivity and an expert audit kit

Which labels are human and which are automatic.

Cohort	Human annotation	Automatic annotation (model or rules)
ISIC 2019 (hair)	diagnoses (ISIC); lesion masks for the 10,015 HAM10000 images (manual)	lesion masks for the other 15,316 images (U-Net trained on manual ISIC 2018 masks); hair/ruler masks (published semi-automatic masks)
TN3K/TNCD (thyroid)	nodule masks; benign/malignant from cytological biopsy	caliper masks for all 3,493 images (rule-based detector)
MMOTU (ovary)	tumour masks; tumour categories	caliper masks for all 1,202 images (same detector)
SEE-AI (capsule)	lesion boxes and classes; contamination masks for 22 frames (expert)	contamination masks for 5,459 of 5,481 frames (linear probe on DINOv2 patch tokens, IoU 0.62)
NIH ChestX-ray14	drain labels of pneumothorax images (NEATX, non-expert annotators)	disease labels (text-mined from reports); lung masks (TorchXRayVision); drain labels of pneumothorax-negative images (drain detector)
All cohorts	—	visual audits of the automatic labels (by the analysis team, which includes an AI agent; no clinician)

Visual audit of 40 random images per cell by the analysis team (not clinicians): thyroid detections 95% on real calipers; caliper-free thyroid images 95–97% without visible calipers; ovary 40/40 detections are overlays, 31/40 calipers; caliper-free ovary images 92% without calipers but $\approx 28\%$ with other overlays. Capsule debris probe: pixel accuracy 0.92, IoU 0.62 against expert masks. Re-running with large-marker (≥ 50 px) and strict-location ($r \geq 0.7$ / $r < 0.05$) definitions (thyroid, ovary) and a strict capsule Trap B leaves every primary conclusion unchanged (SENSITIVITY_ARTIFACT_LABELS). *Expert audit (prepared, not yet carried out)*. `scripts/audit/make_expert_audit_kit.py` draws 60 images per cohort (half flagged by the automatic label), renders each blind (ROI contour only) and with the automatic mask, and writes a rating sheet (artifact present; location relative to the ROI; mask quality); `scripts/audit/score_expert_audit.py` computes Cohen’s κ between two independent raters and the precision and recall of the automatic labels against their consensus, with bootstrap CIs.

S5. Secondary method analyses

- *Patch-token suppression (SLAS)*. A linear probe on DINOv2 patch tokens localises real hair from 5 annotated images (patch AUROC 0.93; 0.97 with 50), but dropping flagged tokens improves reversed AUROC only modestly (+0.02 hair, +0.08 thyroid over ROI pooling).
- *Annotation-free pseudo-groups*. An overlay detector trained on synthetic insertion pairs recognises real calipers (AUROC 0.84) but not hair (0.57); pseudo-group balancing adds +0.04 on thyroid only.
- *Erasure combined with DFR*. Adds ≤ 0.014 over masking plus DFR.
- *Split-validation selector*. Admitting DFR as a candidate (trained and scored on disjoint validation halves) fixes the chest-drain case (0.656 vs 0.538) but makes choices noisier elsewhere; neither selector is uniformly best.

S6. Leakage

Holding cohort, features and protocol fixed, replacing the grouped split with random image-level splits (93 near-duplicate pairs across train/test) inflates the measured shortcut gap by 18% (0% overlap) and 10% (100%) and distorts both masking effects ($+0.184 \rightarrow +0.223$; $-0.129 \rightarrow -0.088$).

S7. Backbone scale

Replacing DINOv2 ViT-B/14 (86 M parameters) by ViT-S/14 (21 M) or ViT-L/14 (304 M), with everything else fixed (pre-registered), the location crossover is positive in all nine cohort–size combinations (ViT-S / ViT-B / ViT-L: thyroid +0.247 / +0.239 / +0.263; ovary +0.296 / +0.170 / +0.207; capsule +0.370 /

+0.377 / +0.417; every CI excludes zero), and protected U-MtE beats masking on thyroid and ovary at every size (ViT-S +0.225 [+0.168, +0.276] and +0.103 [+0.074, +0.134]; ViT-L +0.274 [+0.243, +0.306] and +0.060 [+0.037, +0.084]). Larger models exploit the in-ROI shortcut *more*: the ERM correlated-reversed gap grows from 0.537 to 0.622 (thyroid) and from 0.320 to 0.564 (ovary) between ViT-S and ViT-L, so scale does not remove it. On capsule, disease protection is less complete with ViT-L (−0.044 [−0.056, −0.032] versus masking).

S8. Mitigation methods and their results on the traps

The main text summarises the mitigations as a decision guide. This section gives the annotation-free mask-then-erase method (U-MtE), its results on the traps, the baselines and the end-to-end variants. None of these remedies is claimed to improve performance on unaltered test data (main text).

S8.1 Method: mask-then-erase with a generic insertion library

Insertion pairs. Removing an in-ROI artifact requires its pixels; inserting one does not. Given a frozen encoder f and an overlay generator $\mathcal{G}$, every training image x_i yields a counterfactual $x_i^+ = \mathcal{G}(x_i; \xi_i)$ with random parameters ξ_i and random position, half of the elements centred inside the ROI. Because insertion ignores the label, the differences $\Delta_i = f(x_i^+) - f(x_i)$ are independent of Y .

Generic library. $\mathcal{G}$ is one procedural family used unchanged for every modality and artifact: thin curvilinear strokes, thick bands, solid or translucent patches, blobs, dashed lines and small glyphs, with random colour (half greyscale), opacity and width. It contains no ruler, hair, caliper or debris template; it is a prior over “foreign overlays”, not a model of any particular artifact.

Erasure. The insertion eraser removes the span of the differences:

$$\Sigma_\Delta = \frac{1}{n} \sum_i \Delta_i \Delta_i^\top, \quad U_k = \text{top-}k \text{ eigvecs}(\Sigma_\Delta), \quad P(z) = z - U_k U_k^\top (z - \mu), \quad (1)$$

with k chosen *without labels* as the smallest rank whose held-out paired-difference energy $\mathbb{E}\|P\Delta\|^2 / \mathbb{E}\|\Delta\|^2$ falls below 0.10 (fixed before any experiment).

Mask-then-erase (U-MtE). Masking and erasure have complementary reach: masking removes any artifact outside the ROI but none inside; erasure acts on the representation regardless of location. U-MtE therefore computes the pairs *on masked images* ($\Delta_i = f(m(x_i^+)) - f(m(x_i))$) so that the erased subspace targets exactly what masking leaves behind, and fits the head on $P(f(m(x)))$. When image-level artifact labels exist, the head is group-balanced (U-MtE_{bal}).

Table 1: In-ROI artifacts (Trap A): reversed-test AUROC, with min(reversed, correlated) in parentheses (penalises shortcut flipping). Best robust value per column in bold. Generated by `scripts/make_main_table.py`.

Method	Needs	Hair/DINOv2	Hair/DermLIP	Thyroid/DINOv2	Thyroid/MedSigLIP	Thyroid/ConvNeXt	Ovary/DINOv2	Capsule/DINOv2	Capsule/MedSigLIP	Capsule/ConvNeXt
ERM	—	0.51 (0.51)	0.67 (0.67)	0.29 (0.29)	0.29 (0.29)	0.36 (0.36)	0.41 (0.41)	0.58 (0.58)	0.57 (0.57)	0.53 (0.53)
ROI masking	ROI	0.46 (0.46)	0.56 (0.56)	0.40 (0.40)	0.34 (0.34)	0.35 (0.35)	0.55 (0.55)	0.67 (0.67)	0.69 (0.69)	0.61 (0.61)
Mask + overlay augmentation	ROI	0.47 (0.47)	—	0.41 (0.41)	0.36 (0.36)	0.36 (0.36)	0.53 (0.53)	0.68 (0.68)	0.71 (0.71)	0.63 (0.63)
U-MtE (ours)	ROI	0.50 (0.50)	0.50 (0.50)	0.62 (0.62)	0.59 (0.59)	0.49 (0.49)	0.56 (0.56)	0.54 (0.54)	0.46 (0.46)	0.54 (0.54)
U-MtE, protected (ours)	ROI + Y of $A=0$	0.50 (0.50)	0.59 (0.59)	0.68 (0.68)	0.65 (0.65)	0.54 (0.54)	0.59 (0.59)	0.66 (0.66)	0.66 (0.66)	0.64 (0.64)
Group-balanced	A	0.72 (0.72)	0.81 (0.81)	0.68 (0.68)	0.70 (0.70)	0.68 (0.68)	0.70 (0.70)	0.76 (0.76)	0.80 (0.80)	0.72 (0.72)
DFR	A	0.71 (0.71)	0.78 (0.78)	0.75 (0.67)	0.75 (0.72)	0.73 (0.67)	0.72 (0.69)	0.82 (0.82)	0.82 (0.82)	0.78 (0.78)
Mask + DFR	ROI + A	0.66 (0.66)	—	0.83 (0.75)	0.81 (0.77)	0.76 (0.71)	0.75 (0.75)	0.89 (0.89)	0.91 (0.91)	0.86 (0.86)
U-MtE + balanced (ours)	ROI + A	0.69 (0.69)	0.63 (0.63)	0.79 (0.79)	0.79 (0.79)	0.75 (0.75)	0.73 (0.73)	0.84 (0.84)	0.84 (0.84)	0.80 (0.80)

Disease protection. If the artifact resembles the pathology (intestinal debris versus erosion fibrin), U_k can contain disease evidence. Let W be an orthonormal basis of logistic label directions fitted on artifact-free training images; we erase $U'_k = \text{orth}((I - WW^\top)U_k)$ instead, which guarantees $w^\top P(z) = w^\top z$ for every $w \in \text{span}(W)$ (cf. task-preserving erasure, 3).

Scope. U-MtE is a representation-level intervention: it acts on the features of a frozen encoder, the dominant way in which foundation models are deployed on small medical datasets, and needs only the encoder, the ROI mask and the generator. An end-to-end analogue is evaluated as a secondary check (S8.3).

Tool. The released package exposes insertion, erasure and a patch-token artifact probe for any frozen backbone (ViTs and CNNs) through a two-call API.

S8.2 Results on the traps

U-MtE repairs what masking leaves behind. Without any artifact example, artifact mask or artifact label—it uses only the ROI mask that masking already requires—U-MtE improves reversed AUROC over masking in the in-ROI regime of every modality: thyroid calipers +0.224 [+0.195, +0.253] (DINOv2) and +0.246 [+0.224, +0.269] (MedSigLIP); dermoscopic hair +0.040 [+0.024, +0.056]; synthetic capsule debris +0.121 [+0.078, +0.170]; synthetic calipers +0.278 [+0.214, +0.344]; synthetic chest tubes +0.134 [+0.113, +0.156] (DINOv2 / RAD-DINO). With MedSigLIP the unbalanced method *loses* to masking in both controlled sweeps (debris −0.190 [−0.265, −0.120], calipers −0.076 [−0.120, −0.034]) although it wins on real thyroid calipers (+0.246); the balanced variant beats masking at full overlap in all four controlled sweeps (+0.211 to +0.470) and has no significant location interaction in three of them (with MedSigLIP on capsule it gains *more* at full overlap: interaction −0.165 [−0.300, −0.025]). Disease protection turns the MedSigLIP debris failure into a gain (+0.114 [+0.022, +0.238] over masking at full overlap) and the caliper one into a non-significant difference (−0.041 [−0.095, +0.002]).

Sensitivity to the erasure rank. In a pre-registered ablation (Trap A, DINOv2), the thyroid gain over masking is +0.18 to +0.22 for every setting that erases at least 16 directions (held-out energy 0.5–0.99, or fixed $k \in \{16, 64\}$), but only +0.06 to +0.07 with one to four directions: the shift caused by overlays is multi-dimensional. The pre-specified 90% rule reached the rank cap ($k = 64$) in all three cohorts. The protected variant never harms capsule by more than 0.010 and helps ovary at every rank (+0.007 to +0.043), whereas unprotected erasure harms capsule more the more directions it removes, as the overlap term ρ of the linear-Gaussian account (main paper) predicts.

Erasure, not augmentation. Using the same generic overlays as training augmentation (overlaid copies of a random half of the training images, in the masked view) gains at most +0.03 over masking; erasing their subspace instead gains +0.215 [+0.189, +0.244] (thyroid, DINOv2), +0.225 [+0.202, +0.247] (thyroid, MedSigLIP) and +0.035 [+0.017, +0.055] (hair) more than augmentation. Under the stricter near-duplicate grouping (main paper, stricter leakage grouping) the thyroid advantage is unchanged (+0.192) but the small ovary advantage (+0.035) loses significance, so we regard erase > augment as established for thyroid and hair only.

When the artifact resembles the disease. On real capsule debris U-MtE is worse than masking (−0.126 [−0.146, −0.107]): erosions carry yellow-white fibrin that resembles debris, and a subspace estimated from real-debris templates is worse still (−0.395; archived run, not re-estimated). Disease protection removes the failure (−0.006 [−0.018, +0.006] generic; template variant −0.012, not re-estimated) at no cost where erasure already works (protected minus unprotected: thyroid +0.059 [+0.019, +0.096], hair +0.001 [−0.016, +0.017]).

With image-level artifact labels. Adding group balancing, U-MtE_{bal} is the most robust model in thyroid ultrasound by the minimum of correlated and reversed AUROC (0.788 DINOv2, 0.792 MedSigLIP; DFR 0.659 and 0.719); in real capsule endoscopy masking plus last-layer retraining is higher (0.885 versus 0.857); on dermoscopic hair DFR alone is best (0.741) because most of the hair shortcut is not in the hair pixels (removing every hair patch with the oracle mask gains only +0.015). In both controlled sweeps the benefit of U-MtE_{bal} no longer depends on location (interaction +0.021 [−0.019, +0.061] capsule; −0.053 [−0.108, +0.003] thyroid), whereas that of masking does (+0.228, +0.433). The capsule findings replicate with MedSigLIP (crossover +0.335 [+0.286, +0.383]; masking plus DFR 0.901). All thyroid and capsule findings replicate with a CNN (ConvNeXt-Base): crossovers +0.275 and +0.320; U-MtE − mask +0.135 [+0.102, +0.168] and erase − augment +0.128 [+0.093, +0.161] on thyroid.

Label-free and erasure baselines. Among methods that need no artifact labels, U-MtE beats JTT [5] by +0.206 [+0.171, +0.241] (thyroid), +0.125 [+0.063, +0.189] (ovary) and +0.094 (chest drains, RAD-DINO;

archived run, not re-estimated; not with DINOv2, -0.010); JTT on the unmasked view is best on hair. Task-preserving concept removal (SPLINCE; 3), fitted with artifact labels in the whitened, minimal-distortion form of the original method, removes the artifact covariance at a large cost in disease signal (clean AUROC 0.56–0.78 versus 0.69–0.87 for masking) and still flips the shortcut on capsule endoscopy (reversed 0.868, correlated 0.683): when artifact and label are strongly associated in training, removing one removes much of the other. By $\min(\text{reversed, correlated})$ U-MtE_{bal} exceeds SPLINCE on thyroid, capsule and hair (0.788 vs 0.596; 0.857 vs 0.683; 0.704 vs 0.564); on chest drains SPLINCE is higher (0.609 vs 0.460) but below DFR (0.683).

Text-prompted erasure. With vision–language backbones an artifact direction can be obtained from text alone (“a thyroid ultrasound image with measurement calipers” minus “a thyroid ultrasound image”; biased-prompt construction, 1). Erasing that subspace after masking is a near no-op in every cohort ($+0.001$ to $+0.008$ over masking; MedSigLIP thyroid, ovary, capsule and DermLIP hair), whereas U-MtE with the same MedSigLIP features gains $+0.246$ [$+0.224, +0.269$] (thyroid) and $+0.210$ [$+0.167, +0.253$] (ovary). The direction along which a described artifact moves the *text* embedding is not the direction along which the real artifact moves the *image* embedding; synthetic insertions measure the latter directly.

A single procedure: validation-driven selection. Selecting, per fold, the candidate with the best group-aware validation score (image-level artifact labels needed on validation images only) beats masking in all five cohorts ($+0.17$ to $+0.40$), is within 0.02 of the best candidate in each, and beats DFR in three (thyroid $+0.057$, capsule $+0.047$, ovary $+0.010$); it chooses U-MtE_{bal} where the artifact is an overlay and plain balancing on hair and drains, where DFR remains better. A split-validation variant that admits DFR as a candidate is reported in the supplement.

Modern inpainting. Inpainting the detected caliper pixels with LaMa [6] and then masking is a strong baseline when artifact masks are available (thyroid $+0.226$, ovary $+0.123$ over masking; archived LaMa runs, intervals not re-estimated because their per-image predictions were not kept). U-MtE_{protect}, which needs no artifact mask, matches it on thyroid (-0.009 , not re-estimated) and is below it on ovary (-0.080) and on dermoscopic hair, where published hair masks are available (-0.086); with image-level artifact labels, U-MtE_{bal} exceeds it in all three ($+0.162, +0.067, +0.123$).

S8.3 End-to-end fine-tuning and erase-while-fine-tuning

Fine-tuning ResNet-50 end-to-end (five folds) reproduces the frozen-backbone findings. Capsule: masking harms in-ROI (-0.279 [$-0.377, -0.185$]) and helps out-of-ROI ($+0.301$), crossover $+0.606$ [$+0.495, +0.716$]. Thyroid: the crossover points the same way ($+0.063$ [$-0.022, +0.144$], five folds), and an end-to-end analogue of U-MtE—an invariance penalty between masked images with and without a generic overlay—improves on

masking by +0.139 [+0.096, +0.184] without shortcut flipping (reversed 0.679, correlated 0.703), matching the best label-using arm. It fails on capsule (−0.021), where the artifact resembles the pathology. Ovary (1,202 images) is underpowered: U-MtE end-to-end has the best reversed, correlated and clean AUROC, with CIs overlapping masking. A fine-tuned ViT-S reproduces the crossover on thyroid (+0.122 [+0.039, +0.211]), but the end-to-end U-MtE analogue does not improve on masking with this architecture (−0.000 [−0.026, +0.026]). Applying U-MtE post hoc to the penultimate features of the masked, fine-tuned ResNet-50 does not help either (pre-registered; −0.000 [−0.001, +0.001] over the same features without erasure): after task training the shortcut is encoded along task-specific directions that generic overlays do not move. For fully fine-tuned networks the subspace must be estimated and removed *during* training. *Erase-while-fine-tuning* (pre-registered; no artifact annotation) projects a running estimate of the overlay-shift subspace of the network’s own penultimate features out before the head at every step and at test time, and adds feature invariance and prediction consistency between masked images with and without a generic overlay. On thyroid it improves on masking by +0.195 [+0.153, +0.237] with ResNet-50 and +0.119 [+0.036, +0.198] with ViT-S (reversed AUROC 0.693 and 0.608 vs 0.536 and 0.465), whereas each component alone works for only one architecture (projection + invariance: ResNet-50 +0.186 [+0.144, +0.230], ViT-S +0.032 [−0.093, +0.144]; prediction consistency: ResNet-50 +0.042 [+0.011, +0.074], ViT-S +0.061 [+0.036, +0.084]; with the corrected bootstrap consistency alone is now significant for both, but smaller than the combination). With ResNet-50 the correlated AUROC falls slightly below the reversed one (0.661 vs 0.693), a mild over-correction. On ovary (ResNet-50) it is positive but not significant (+0.045 [−0.036, +0.130]; every fine-tuning contrast on this 1,202-image cohort is underpowered), and on capsule, where debris resembles erosion fibrin, it fails (−0.026 [−0.056, +0.006]), as does every annotation-free method.

S9. Operating points on every unaltered test set

Masking minus ERM at the five pre-registered validation-fixed operating points (95% hierarchical bootstrap CIs; thresholds re-estimated in every replicate). *Conflicting*: positive cases whose in-ROI artifact status points to the negative class. Generated by `scripts/analysis/operating_points.py`.

Test set	Threshold (validation)	Δ sensitivity	Δ sensitivity, conflicting	Δ specificity
Thyroid	maxBA	-0.275 [-0.435, -0.138]	-0.479 [-0.640, -0.313]	+0.202 [+0.115, +0.330]
	spec0.80	-0.202 [-0.300, -0.100]	-0.372 [-0.499, -0.236]	+0.117 [+0.059, +0.172]
	spec0.90	-0.152 [-0.253, -0.043]	-0.236 [-0.383, -0.089]	+0.075 [+0.029, +0.122]
	sens0.80	-0.373 [-0.469, -0.270]	-0.546 [-0.671, -0.421]	+0.278 [+0.203, +0.359]
	sens0.90	-0.281 [-0.391, -0.178]	-0.462 [-0.614, -0.310]	+0.339 [+0.236, +0.425]
ISIC, BCN held out	maxBA	+0.000 [-0.044, +0.055]	+0.001 [-0.045, +0.059]	-0.078 [-0.174, +0.013]
	spec0.80	+0.017 [-0.003, +0.036]	+0.019 [-0.004, +0.042]	-0.110 [-0.135, -0.079]
	spec0.90	+0.042 [+0.012, +0.071]	+0.042 [+0.006, +0.076]	-0.104 [-0.135, -0.074]
	sens0.80	+0.002 [-0.029, +0.035]	+0.003 [-0.030, +0.038]	-0.082 [-0.138, -0.020]
	sens0.90	+0.001 [-0.029, +0.030]	-0.001 [-0.036, +0.034]	-0.065 [-0.134, +0.016]
ISIC, HAM held out	maxBA	-0.006 [-0.117, +0.150]	+0.002 [-0.110, +0.165]	+0.042 [-0.070, +0.124]
	spec0.80	-0.062 [-0.110, -0.015]	-0.055 [-0.104, -0.007]	+0.092 [+0.064, +0.121]
	spec0.90	-0.057 [-0.115, -0.005]	-0.053 [-0.113, +0.004]	+0.041 [+0.020, +0.066]
	sens0.80	-0.004 [-0.055, +0.044]	+0.009 [-0.049, +0.062]	+0.061 [+0.017, +0.114]
	sens0.90	-0.021 [-0.060, +0.017]	-0.013 [-0.056, +0.031]	+0.096 [+0.041, +0.160]
ISIC, MSK held out	maxBA	-0.013 [-0.108, +0.080]	-0.033 [-0.129, +0.062]	+0.032 [-0.033, +0.104]
	spec0.80	-0.074 [-0.127, -0.026]	-0.094 [-0.149, -0.044]	+0.083 [+0.064, +0.110]
	spec0.90	-0.091 [-0.145, -0.043]	-0.110 [-0.167, -0.058]	+0.058 [+0.042, +0.075]
	sens0.80	-0.050 [-0.105, +0.000]	-0.070 [-0.129, -0.017]	+0.068 [+0.037, +0.105]
	sens0.90	-0.023 [-0.080, +0.029]	-0.037 [-0.096, +0.017]	+0.073 [+0.024, +0.124]
Capsule	maxBA	+0.086 [+0.017, +0.169]	-0.074 [-0.167, +0.048]	+0.089 [+0.031, +0.152]
	spec0.80	+0.140 [+0.082, +0.222]	+0.022 [-0.041, +0.123]	+0.041 [-0.029, +0.107]
	spec0.90	+0.245 [+0.159, +0.316]	+0.092 [-0.039, +0.214]	-0.000 [-0.039, +0.057]
	sens0.80	+0.030 [-0.014, +0.080]	-0.096 [-0.201, -0.006]	+0.092 [+0.045, +0.148]
	sens0.90	+0.022 [-0.026, +0.072]	-0.104 [-0.204, -0.026]	+0.169 [+0.092, +0.275]

S10. Absolute performance and published results

Only the thyroid cohort has a published benchmark on the same split and labels (official patient-disjoint TNCD split, 614 test images). The ovarian (benign cystic versus solid-component tumours) and capsule (erosion versus polyp-like) binary tasks and the ISIC 2019 leave-one-hospital-out melanoma task were defined for this study and have no published counterpart; their clean AUROC is reported with every result table. Frozen encoders with a linear head are below fine-tuned CNNs; the location law replicates with fine-tuned networks (main text).

Model	Source	Test AUROC
ResNet-18, cross-entropy (fine-tuned)	Gong et al. [2]	0.773
DenseNet-121, cross-entropy (fine-tuned)	Gong et al. [2]	0.780
ConvNeXt-T, cross-entropy (fine-tuned)	Gong et al. [2]	0.784
ResNet-18, adaptive curriculum (fine-tuned)	Gong et al. [2]	0.799
DINOv2 ViT-B/14 frozen + linear head, ERM (rebuilt run)	this work	0.735

Table 2: Covariate balance of the location traps: standardised mean differences (SMD) between the artifact-bearing images of Trap A (in-ROI) and Trap B (out-of-ROI), before and after propensity matching (categorical covariates: largest level). Positive: larger in Trap A.

Cohort	Covariate	SMD before	SMD after
ISIC hair	lesion area	+2.13	+0.03
	hair amount	-0.16	+0.11
	age	+0.14	-0.01
	age missing	-0.04	-0.05
	sex (missing)	-0.03	-0.04
	site (torso)	+0.19	-0.01
	benign dx (MEL)	+0.26	+0.00
	source (MSK)	-0.54	+0.00
Thyroid	roi area	+0.40	-0.01
	caliper px log	+0.53	+0.04
	width	-0.14	+0.04
	height	-0.12	+0.02
	mean grey	+0.18	-0.01
	roi centroid row	+0.06	+0.01
Ovary	roi area	+0.32	-0.00
	caliper px log	+0.39	+0.03
	width	-0.30	+0.04
	height	+0.05	+0.00
	mean grey	+0.11	+0.02
	subtype (2.0)	-0.16	-0.01
Capsule	roi area	+2.57	+0.01
	contamination	-0.33	-0.12
	mean grey	+0.13	+0.11
	frame block (0.0)	-0.37	-0.23

S11. Covariate balance of the traps and the real-artifact transplant

Full matched analysis (pre-registered calliper 0.2 SD, and 0.05 SD as a sensitivity analysis).

Cohort	Calliper (SD)	In-ROI / out-of-ROI images	Kept per trap (neg./pos.)	SMD _{max} before	after (pooled)	after (within label)	Crossover unmatched	Crossover matched
ISIC hair	0.20	5,438 / 8,329	1,829 (1431/398)	2.13	0.11	0.16	+0.147 [+0.111, +0.185]	+0.154 [+0.123, +0.186]
	0.05	5,438 / 8,329	1,777 (1394/383)	2.13	0.13	0.18	+0.147 [+0.111, +0.185]	+0.124 [+0.086, +0.162]
Thyroid	0.20	1,150 / 326	316 (223/93)	0.53	0.04	0.15	+0.239 [+0.196, +0.280]	+0.238 [+0.186, +0.290]
	0.05	1,150 / 326	299 (212/87)	0.53	0.06	0.19	+0.239 [+0.196, +0.280]	+0.218 [+0.165, +0.270]
Ovary	0.20	617 / 194	179 (86/93)	0.39	0.08	0.14	+0.170 [+0.087, +0.252]	+0.163 [+0.050, +0.273]
	0.05	617 / 194	165 (82/83)	0.39	0.08	0.18	+0.170 [+0.087, +0.252]	+0.152 [+0.041, +0.265]
Capsule	0.20	816 / 1,240	71 (19/52) [†]	2.57	0.23	0.49	+0.377 [+0.327, +0.426]	+0.230 [+0.036, +0.434]
	0.05	816 / 1,240	56 (11/45) [†]	2.57	0.23	0.67	+0.377 [+0.327, +0.426]	+0.328 [+0.087, +0.573]

[†]Infeasible (fewer than 30 pairs in one label). Crossed seed×image CIs. *Paste-edge control.* Neutral tissue from the donor image (or, where no artifact-free window exists, from an artifact-free image of the cohort) pasted through the same masks, transforms, positions and blending as the artifact.

Table 3: Real-artifact transplant: reversed / correlated / clean AUROC of ERM and ROI masking with the same real artifact inside or outside the ROI of the same images, and the change from masking (95% CI). $|\Delta p|$: same-head counterfactual sensitivity to inserting the artifact.

Cohort	Location	ERM rev / corr / clean	Mask rev / corr / clean	Mask – ERM (rev)	$ \Delta p $ ERM / mask
ISIC hair	outside ROI	0.539 / 0.869 / 0.759	0.716 / 0.716 / 0.716	+0.178 [+0.128, +0.229]	0.230 / 0.000
	inside ROI	0.454 / 0.904 / 0.776	0.379 / 0.842 / 0.672	−0.075 [−0.115, −0.036]	0.340 / 0.297
Thyroid	outside ROI	0.497 / 0.857 / 0.708	0.846 / 0.846 / 0.846	+0.349 [+0.321, +0.378]	0.201 / 0.000
	inside ROI	0.487 / 0.851 / 0.708	0.559 / 0.906 / 0.801	+0.072 [+0.044, +0.100]	0.206 / 0.271
Ovary	outside ROI	0.686 / 0.759 / 0.725	0.750 / 0.750 / 0.750	+0.064 [+0.002, +0.127]	0.023 / 0.000
	inside ROI	0.657 / 0.771 / 0.723	0.647 / 0.825 / 0.755	−0.010 [−0.075, +0.057]	0.037 / 0.116
Capsule	outside ROI	0.592 / 0.905 / 0.795	0.914 / 0.914 / 0.914	+0.322 [+0.264, +0.383]	0.193 / 0.000
	inside ROI	0.508 / 0.906 / 0.787	0.333 / 0.956 / 0.871	−0.175 [−0.241, −0.108]	0.238 / 0.551

Cohort	Artifact	Neutral paste	Neutral / artifact	Artifact – neutral	p_{Holm}	Attribu
ISIC hair	+0.253 [+0.214, +0.293]	+0.204 [+0.164, +0.244]	0.81	+0.049 [+0.026, +0.072]	0.0004	
Thyroid	+0.277 [+0.256, +0.298]	+0.145 [+0.130, +0.161]	0.52	+0.132 [+0.116, +0.148]	0.0004	
Ovary	+0.074 [+0.044, +0.104]	+0.052 [+0.021, +0.084]	0.71	+0.022 [−0.002, +0.048]	0.1372	
Capsule	+0.497 [+0.427, +0.567]	+0.459 [+0.383, +0.535]	0.92	+0.037 [−0.017, +0.093]	0.1792	

Pre-registered rule: attributed to the artifact only if artifact – neutral > 0 (Holm) and the neutral interaction is below one quarter of the artifact’s. *Transplant design.* Recipients: artifact-free images of each cohort. Donors: artifact-bearing images of the same cohort; an artifact instance is the largest connected component of its mask after a grouping dilation (calipers 9px, hair and debris 5px), with at least 30 pixels and a bounding-box side of at most 259px. For every recipient, donors are tried in a stable random order (up to 50 donors $\times$ 8 dihedral transforms) until one fits both inside (overlap ≥ 0.95) and outside (overlap 0) the ROI within the field of view; the instance is alpha-composited with a Gaussian-blurred mask. Environments as in the controlled sweeps (0.9/0.1 training and correlated test, 0.1/0.9 reversed test, clean test without artifact), 5-fold grouped cross-validation, five presence seeds (bootstrap clusters). Recipients without a feasible donor (mostly small capsule lesion boxes) are excluded; counts by label are in `results/review2/transplant/*/feasibility_by_label.csv`.

S15. Per-seed versus crossed bootstrap

The original estimator resamples training-seed clusters and then test images independently within each sampled cluster. When the clusters share test images—every seed of an analysis scores the same official thyroid test split, the same transplant recipients, and largely the same trap pool—

this treats one test set as several and understates the test-set sampling variance. The crossed estimator resamples clusters and draws one Poisson weight per test image, shared by every cluster (`wtss.stats.crossed_auc_bootstrap`). For the claims below both CIs are shown; the paper uses the crossed ones. Generated by `scripts/analysis/crossed_ci.py`.

Claim	Cohort / test set	Crossed seed×image bootstrap	Width relative to the original	Verdict changed
real-trap crossover (repro)	ISIC hair	+0.147 [+0.109, +0.185]	1.01	no
real-trap crossover (matched)	ISIC hair	+0.154 [+0.122, +0.187]	1.03	no
real-trap crossover (matched05)	ISIC hair	+0.124 [+0.086, +0.162]	1.00	no
real-trap crossover (repro)	Thyroid	+0.239 [+0.196, +0.280]	1.00	no
real-trap crossover (matched)	Thyroid	+0.238 [+0.186, +0.291]	1.02	no
real-trap crossover (matched05)	Thyroid	+0.218 [+0.165, +0.270]	1.00	no
real-trap crossover (repro)	Ovary	+0.170 [+0.087, +0.253]	1.00	no
real-trap crossover (matched)	Ovary	+0.163 [+0.049, +0.274]	1.00	no
real-trap crossover (matched05)	Ovary	+0.152 [+0.041, +0.265]	0.98	no
real-trap crossover (repro)	Capsule	+0.377 [+0.326, +0.425]	1.00	no
real-trap crossover (matched)	Capsule	+0.230 [+0.029, +0.422]	0.99	no
real-trap crossover (matched05)	Capsule	+0.328 [+0.087, +0.573]	1.01	no
transplant interaction (artifact)	ISIC hair	+0.253 [+0.214, +0.293]	1.00	no
transplant interaction (balanced)	ISIC hair	-0.052 [-0.071, -0.033]	1.00	no
transplant interaction (neutral paste)	ISIC hair	+0.204 [+0.164, +0.244]	1.01	no
transplant interaction (artifact)	Thyroid	+0.277 [+0.256, +0.298]	0.97	no
transplant interaction (balanced)	Thyroid	-0.016 [-0.025, -0.007]	0.99	no
transplant interaction (neutral paste)	Thyroid	+0.145 [+0.130, +0.161]	1.03	no
transplant interaction (artifact)	Ovary	+0.074 [+0.044, +0.104]	0.98	no
transplant interaction (balanced)	Ovary	-0.011 [-0.025, +0.002]*	0.98	no
transplant interaction (neutral paste)	Ovary	+0.052 [+0.021, +0.084]	0.99	no
transplant interaction (artifact)	Capsule	+0.497 [+0.427, +0.567]	1.01	no
transplant interaction (balanced)	Capsule	-0.032 [-0.068, +0.004]*	1.00	no
transplant interaction (neutral paste)	Capsule	+0.459 [+0.383, +0.535]	1.01	no
natural mte - mask (all pairs)	Thyroid (official split)	+0.004 [-0.012, +0.020]*	1.00	no
natural mte_protect - mask (all pairs)	Thyroid (official split)	-0.021 [-0.034, -0.008]	0.95	no
natural balanced - mask (all pairs)	Thyroid (official split)	-0.017 [-0.064, +0.030]*	0.98	no
natural mte_balanced - mask (all pairs)	Thyroid (official split)	-0.005 [-0.023, +0.013]*	1.01	no
natural mask - ERM (hard pairs)	Thyroid (official split)	-0.095 [-0.167, -0.023]	1.00	no
natural mask - ERM (all pairs)	Thyroid (official split)	-0.004 [-0.048, +0.041]*	1.00	no
natural mte - mask (all pairs)	ISIC, BCN held out	+0.007 [+0.004, +0.010]	0.90	no
natural mte_protect - mask (all pairs)	ISIC, BCN held out	-0.019 [-0.024, -0.014]	0.94	no
natural balanced - mask (all pairs)	ISIC, BCN held out	+0.022 [+0.012, +0.031]	0.97	no
natural mte_balanced - mask (all pairs)	ISIC, BCN held out	+0.010 [+0.006, +0.015]	0.93	no
natural mask - ERM (hard pairs)	ISIC, BCN held out	-0.025 [-0.042, -0.008]	0.99	no
natural mask - ERM (all pairs)	ISIC, BCN held out	-0.025 [-0.035, -0.016]	0.97	no
natural mte - mask (all pairs)	ISIC, HAM held out	+0.013 [+0.009, +0.017]	1.04	no
natural mte_protect - mask (all pairs)	ISIC, HAM held out	-0.006 [-0.016, +0.004]*	1.00	no
natural balanced - mask (all pairs)	ISIC, HAM held out	-0.032 [-0.049, -0.013]	0.99	no
natural mte_balanced - mask (all pairs)	ISIC, HAM held out	+0.018 [+0.009, +0.028]	1.03	no
natural mask - ERM (hard pairs)	ISIC, HAM held out	+0.050 [+0.030, +0.070]	1.00	no
natural mask - ERM (all pairs)	ISIC, HAM held out	+0.031 [+0.017, +0.045]	0.99	no
natural mte - mask (all pairs)	ISIC, MSK held out	-0.008 [-0.014, -0.002]	1.02	no
natural mte_protect - mask (all pairs)	ISIC, MSK held out	-0.003 [-0.012, +0.006]*	1.03	no
natural balanced - mask (all pairs)	ISIC, MSK held out	-0.018 [-0.041, +0.004]*	0.98	no
natural mte_balanced - mask (all pairs)	ISIC, MSK held out	-0.016 [-0.024, -0.008]	1.03	no
natural mask - ERM (hard pairs)	ISIC, MSK held out	-0.065 [-0.111, -0.020]	0.99	no
natural mask - ERM (all pairs)	ISIC, MSK held out	+0.016 [-0.006, +0.038]*	0.98	no
natural mte - mask (all pairs)	Capsule (held-out frames)	-0.005 [-0.011, +0.001]*	0.98	no
natural mte_protect - mask (all pairs)	Capsule (held-out frames)	-0.023 [-0.029, -0.017]	0.94	no
natural balanced - mask (all pairs)	Capsule (held-out frames)	-0.086 [-0.109, -0.065]	0.98	no
natural mte_balanced - mask (all pairs)	Capsule (held-out frames)	-0.008 [-0.013, -0.001]	1.08	no
natural mask - ERM (hard pairs)	Capsule (held-out frames)	+0.021 [-0.003, +0.048]*	1.00	no
natural mask - ERM (all pairs)	Capsule (held-out frames)	+0.077 [+0.059, +0.097]	0.99	no
fine-tuned crossover (ResNet-50)	ISIC hair	+0.143 [+0.056, +0.231]	0.98	no

*Crossed CI includes zero. The original intervals are listed in `results/bootstrap_correction/sweeps_umte_old_vs_new.md`.

S16. Later registered analyses: implementations, overlap, external natural test, labels, fine-tuning, thresholds

Four registrations added after the main analyses (PREREGISTRATION_ROUND4 to ROUND7; registry in S1), each committed and pushed before its analysis; the hypothesis identifiers below (FT0–FT4, TM1–TM2) are those of the registrations. Full reports: `stages/stage7_robustness_external` and `stages/stage8_labels_strong_model`.

Masking implementations and the real overlap. Five views per image from the cached 518-px image and its ROI: mean-colour fill (the main analysis), black fill, a 16-px Gaussian blur of everything outside the ROI, a crop to the ROI’s bounding box, and the same crop of the masked image. For the dose-response, all artifact-bearing images of a cohort are binned by r (registered bins, merged by a registered rule when a bin is too small); the slope of the mask gain on the mean r of each bin is estimated with the crossed bootstrap.

Table 4: Location crossover C_v for every masking implementation (DINOv2 ViT-B/14, crossed 95% CIs). All 16 non-reference crossovers are positive with Holm-adjusted $p \leq 0.010$ (family of 16).

Cohort	mean-colour fill (reference)	black fill	blurred background	box crop	box crop of the masked image
Dermoscopy hair (ISIC 2019)	+0.147 [+0.111, +0.185]	+0.170 [+0.133, +0.207]	+0.154 [+0.117, +0.191]	+0.091 [+0.064, +0.117]	+0.154 [+0.121, +0.187]
Thyroid calipers	+0.239 [+0.196, +0.280]	+0.223 [+0.181, +0.265]	+0.218 [+0.177, +0.258]	+0.246 [+0.206, +0.285]	+0.250 [+0.210, +0.290]
Capsule debris	+0.377 [+0.328, +0.426]	+0.367 [+0.318, +0.416]	+0.278 [+0.221, +0.330]	+0.317 [+0.267, +0.367]	+0.358 [+0.305, +0.409]
Ovary calipers	+0.170 [+0.088, +0.250]	+0.153 [+0.063, +0.242]	+0.112 [+0.027, +0.195]	+0.145 [+0.083, +0.207]	+0.160 [+0.080, +0.240]

Table 5: Secondary contrast $C_{\text{mask}} - C_v$: how much smaller the crossover of an implementation is than that of mean-colour fill (positive = weaker location effect than the reference).

Cohort	black fill	blurred background	box crop	box crop of the masked image
Dermoscopy hair (ISIC 2019)	−0.023 [−0.040, −0.007]	−0.006 [−0.042, +0.024]	+0.056 [+0.012, +0.099]	−0.007 [−0.033, +0.019]
Thyroid calipers	+0.016 [−0.004, +0.036]	+0.020 [−0.008, +0.049]	−0.007 [−0.049, +0.032]	−0.011 [−0.037, +0.014]
Capsule debris	+0.010 [−0.001, +0.021]	+0.099 [+0.067, +0.132]	+0.059 [+0.032, +0.087]	+0.019 [+0.001, +0.037]
Ovary calipers	+0.017 [−0.034, +0.079]	+0.058 [+0.004, +0.113]	+0.024 [−0.045, +0.096]	+0.010 [−0.046, +0.068]

Table 6: Dose-response: least-squares slope of $g_b = \text{rev}(\text{mask}) - \text{rev}(\text{ERM})$ on the bin mean overlap r (crossed 95% CIs; the artifact-free images shared by all bins carry one bootstrap weight).

Cohort	Bins	Slope of the mask gain on r	Holm p	Spearman	Zero crossing r	H9
Dermoscopy hair (ISIC 2019)	5	−0.235 [−0.276, −0.194]	< 0.001	−1.00	0.401	supported
Thyroid calipers	5	−0.250 [−0.303, −0.197]	< 0.001	−1.00	none	supported
Capsule debris	5	−0.489 [−0.539, −0.438]	< 0.001	−1.00	none	supported
Ovary calipers	3	−0.205 [−0.300, −0.109]	< 0.001	−1.00	none	supported

Prospective test of the linear-Gaussian account. The procedure of the main text (linear-Gaussian account), unchanged, applied to cells that did not exist when it was registered.

Table 7: Prospective theory test: fitted to each cell’s clean and correlated AUROC only; the reversed AUROC and every quantity derived from it are held out. None of these cells existed when the rule was registered.

Endpoint	Cells	Theory	“reversed = clean”	Symmetric reversal
Reversed AUROC, all new cells	84	MAE 0.011, $r = 0.996$	MAE 0.177	MAE 0.036
Masking-implementation crossovers	16	signs 16/16, $r = 0.992$, MAE 0.011		
Overlap dose-response slopes	4	signs 4/4, MAE 0.018		
Overlap-bin gains	18	signs 18/18, MAE 0.014		

ISIC 2019 $\rightarrow$ ISIC 2020 at natural prevalence. All ISIC 2019 images for training and validation; every ISIC 2020 image with a lesion mask for testing, with near-duplicates of ISIC 2019 images removed. The registered rule (perceptual-hash distance ≤ 8) removed 23% of ISIC 2020, mostly look-alike images of other patients; a label-free calibrated rule (hash and embedding thresholds set from the false-match rate between different patients) was registered as a sensitivity analysis before any model of it was fitted.

Table 8: ISIC 2020 under four duplicate rules, from one set of predictions for all 32,997 images (crossed 95% CIs). The calibrated rule is label-free and was committed before any model of this analysis was fitted.

Duplicate rule	Removed	Test images	Melanomas	mask – ERM, all pairs	hard pairs	easy pairs	balanced – mask, hard pairs
exact copies only	26	32,971	578	+0.017 [−0.000, +0.033]	−0.037 [−0.061, −0.012]	+0.048 [+0.009, +0.090]	+0.072 [+0.048, +0.096]
hash distance ≤ 2	138	32,859	575	+0.016 [−0.000, +0.032]	−0.037 [−0.062, −0.013]	+0.048 [+0.008, +0.090]	+0.072 [+0.048, +0.097]
calibrated (primary)	45	32,952	575	+0.016 [−0.000, +0.033]	−0.037 [−0.061, −0.012]	+0.048 [+0.009, +0.090]	+0.072 [+0.048, +0.097]
hash distance ≤ 8 (registered)	7,657	25,340	483	+0.020 [+0.003, +0.038]	−0.026 [−0.052, −0.000]	+0.045 [+0.005, +0.087]	+0.059 [+0.035, +0.084]

Table 9: Cleaner hair groups on the calibrated ISIC 2020 test set: hair-free ($\leq \tau$ predicted hair pixels), clear in-lesion hair ($\geq 3\tau$, $r \geq 0.5$) and clear hair beside the lesion only ($\geq 3\tau$, $r < 0.1$); ambiguous images excluded from these pairs.

Positives	Negatives	Contrast	AUROC difference [95% CI]	Status
hair-free melanomas	benign, clear in-lesion hair	mask – ERM	−0.005 [−0.057, +0.048]	H13
melanomas, clear in-lesion hair	hair-free benign	mask – ERM	+0.022 [−0.019, +0.064]	descriptive
melanomas, hair beside the lesion only	hair-free benign	mask – ERM	−0.028 [−0.060, +0.004]	descriptive
hair-free melanomas	benign, hair beside the lesion only	mask – ERM	+0.059 [+0.010, +0.112]	descriptive
hair-free melanomas	benign, clear in-lesion hair	balanced – mask	+0.016 [−0.039, +0.075]	descriptive

Artifact labels checked without a clinician. Sources, licences and coverage were recorded before any agreement was computed; images on which our own labellers were trained are excluded. A model labeller counts only if its κ is ≥ 0.40 on the images on which the two other caliper labels agree; the annotation detector is declared invalid if it fires on more than half of our caliper-free images. An image is removed if an informative source places it in another cell (artifact present vs absent, or the other side of the boundary); images without a source are kept.

Table 10: Independent label sources and their coverage of our cohorts after the registered exclusions (training images of our own labellers). No image of the thyroid or ovary cohorts left the GPU machine: BUSClean and MedGemma ran locally.

Cohort	Source	Used for		Licence	Matched im- ages	Gate
Dermoscopy hair	DermArtifactDB	hair present / absent		CC BY 4.0	20,519	analysed
Dermoscopy hair	IMA++	lesion masks, 1–5 annota- tors		cc-by-nc-nd-4.0	241	analysed
Dermoscopy hair	Kabir hair masks	hair masks		CC BY 4.0	485	analysed
Capsule debris	Multicentre clear/contaminated capsule masks	contamination masks		CC BY 4.0	29	not feasible
Thyroid calipers	BUSClean	annotation (<code>detect_anno</code>)	detector	MIT	3,493	analysed
Ovary calipers	BUSClean	annotation (<code>detect_anno</code>)	detector	MIT	1,202	analysed
Thyroid calipers	MedGemma 1.5 4B-it	vision–language (presence, location)	answers	model terms of use	3,492	analysed
Ovary calipers	MedGemma 1.5 4B-it	vision–language (presence, location)	answers	model terms of use	1,198	analysed

Table 11: Dermoscopy: agreement of our labels with three independent sources (95% bootstrap CIs over images, 2,000 replicates). The Kabir images are all hair images and our masks mark hair on every one of them, so κ is not informative there; the raw agreement is given instead.

Source	Quantity	Images	All	$Y = 0$ (benign)	$Y = 1$ (melanoma)
DermArtifactDB	hair present (ours: > 30 hair pixels)	20,519	κ 0.076 [0.070, 0.082]	κ 0.083 [0.076, 0.090]	κ 0.051 [0.041, 0.060]
IMA++ (majority mask)	lesion mask (Dice)	241	0.851 [0.830, 0.870]	0.851 [0.825, 0.872]	0.853 [0.819, 0.888]
IMA++	trap class (r recomputed)	239	0.841 [0.791, 0.887]	0.855 [0.807, 0.899]	0.750 [0.594, 0.875]
Kabir et al.	hair mask (Dice)	485	0.618 [0.600, 0.637]	0.617 [0.595, 0.638]	0.629 [0.592, 0.665]
Kabir et al.	hair present	485	agreement 97.3%		

Table 12: Thyroid: how often each independent labeller reports a caliper in each of our cells, and the share of images the registered rule removes (an image is removed if *any* informative labeller places it in another cell).

Our cell	Images	BUSClean: annotation found	MedGemma: “yes”	caliper	Contradicted (regis- tered rule)
caliper-free	1,801	5%	29%		30%
Trap A (in the nodule)	1,150	56%	99%		61%
Trap B (outside)	326	37%	87%		87%

Table 13: Thyroid, **exploratory**: the images on which BUSClean and MedGemma give the same cell. An image counts as contradicted only if both place it in the same cell and that cell differs from ours.

Our cell	Both labellers agree with each other	... and with our cell	Contradicted by both	benign	malignant
caliper-free	1,301	97%	2%	2%	2%
Trap A (in the nodule)	475	94%	2%	3%	1%
Trap B (outside)	89	48%	14%	11%	21%

Table 14: Traps re-run on the images no informative source contradicts (DINOv2 ViT-B/14, protocol of the real traps, crossed 95% CIs, Holm over the cohorts that ran). For capsule and ovary no image was removed, so the re-run reproduces the original analysis and adds no evidence about the labels.

Cohort	Original crossover	Uncontradicted bels	la- Removed (Trap A / Trap B)	Registered test
Dermoscopy hair (ISIC 2019)	+0.147 [+0.111, +0.185]	+0.199 [+0.164, +0.233]	42% / 43%	supported
Thyroid calipers	+0.239 [+0.196, +0.280]	—	61% / 87%	count gate not met (Trap B)
Capsule debris	+0.377 [+0.327, +0.426]	+0.377 [+0.327, +0.425]	0% / 0%	supported; nothing removed (no source analysable)
Ovary calipers	+0.170 [+0.087, +0.252]	+0.170 [+0.089, +0.255]	0% / 0%	supported; nothing removed (no informative source yet)

Fine-tuned thyroid model. Twelve recipes compared on validation images only; the chosen recipe was committed before the final models were trained.

Table 15: Recipe selection on validation images only; no test image was scored. The chosen recipe (bold) was committed before any final model was trained.

Architecture	Learning rate	Epochs	Validation AUROC (ERM, seed 42)
ConvNeXt-T (IN-22k)	3e-5	8	0.849
ConvNeXt-T (IN-22k)	3e-5	20	0.842
ConvNeXt-T (IN-22k)	1e-4	8	0.865
ConvNeXt-T (IN-22k)	1e-4	20	0.865
ResNet-50	3e-5	8	0.658
ResNet-50	3e-5	20	0.723
ResNet-50	1e-4	8	0.723
ResNet-50	1e-4	20	0.781
DINOv2 ViT-B/14	3e-5	8	0.838
DINOv2 ViT-B/14	3e-5	20	0.850
DINOv2 ViT-B/14	1e-4	8	0.715
DINOv2 ViT-B/14	1e-4	20	0.804

Table 16: Part B: the chosen fine-tuned model on the official patient-disjoint thyroid split (5 seeds $\times$ 4 arms) and in the thyroid traps (10 bootstrap clusters). Crossed bootstrap; Holm over FT1–FT4.

	Hypothesis	Estimate [95% CI]	p (one-sided)	Holm p	Verdict
FT0	ERM test AUROC ≥ 0.773 (descriptive)	0.769	—	—	below benchmark
FT1	hard-pair AUROC, mask $-$ ERM < 0	$-0.102 [-0.178, -0.026]$	0.0040	0.0040	supported
FT2	sensitivity loss (CI < 0) at ≥ 3 of OP2–OP5	4 of 4	0.0005	0.0015	supported
FT3	OP2 sensitivity, malignant with in-ROI caliper, mask $-$ ERM < 0	$-0.521 [-0.642, -0.382]$	0.0005	0.0015	supported
FT4	thyroid trap crossover > 0	$+0.187 [+0.111, +0.268]$	0.0001	0.0004	supported

Table 17: Fine-tuned model, official patient-disjoint thyroid test split: masking at five validation-fixed operating points (crossed bootstrap, thresholds re-estimated in every replicate).

Operating point (fixed on validation)	Sensitivity ERM $\rightarrow$ mask	Δ sensitivity	Δ specificity	Sensitivity, malignant with in-ROI caliper (ERM $\rightarrow$ mask)	Δ in that subgroup
OP1 max. balanced accuracy	0.781 $\rightarrow$ 0.408	$-0.373 [-0.508, -0.243]$	$+0.239 [+0.155, +0.345]$	0.779 $\rightarrow$ 0.269	$-0.510 [-0.660, -0.346]$
OP2 specificity ≥ 0.80	0.747 $\rightarrow$ 0.371	$-0.376 [-0.462, -0.297]$	$+0.225 [+0.163, +0.291]$	0.744 $\rightarrow$ 0.223	$-0.521 [-0.642, -0.382]$
OP3 specificity ≥ 0.90	0.542 $\rightarrow$ 0.218	$-0.324 [-0.419, -0.228]$	$+0.143 [+0.098, +0.198]$	0.526 $\rightarrow$ 0.108	$-0.418 [-0.536, -0.276]$
OP4 sensitivity ≥ 0.80	0.783 $\rightarrow$ 0.423	$-0.360 [-0.439, -0.257]$	$+0.232 [+0.159, +0.302]$	0.779 $\rightarrow$ 0.267	$-0.513 [-0.627, -0.371]$
OP5 sensitivity ≥ 0.90	0.888 $\rightarrow$ 0.646	$-0.242 [-0.345, -0.162]$	$+0.240 [+0.159, +0.345]$	0.867 $\rightarrow$ 0.510	$-0.356 [-0.497, -0.225]$

Thresholds matched on the test set. Each arm’s threshold is chosen on the test images themselves, the best case for any re-tuning at a deployment site; the crossed bootstrap re-estimates the thresholds in every replicate.

Table 18: Sensitivity when each arm’s threshold is chosen on the test set itself, so that both arms reach the same test specificity (or sensitivity); ERM $\rightarrow$ mask, and mask $-$ ERM with crossed 95% CIs. The first row of each model is the registered test (TM1, TM2).

Model	Both arms matched at	Malignant, in-ROI caliper (78)	Malignant, no in-ROI caliper	All malignant	Specificity ERM / mask
fine-tuned	test specificity 0.80	0.544 $\rightarrow$ 0.390, $-0.154 [-0.302, -0.023]$	0.563 $\rightarrow$ 0.614, $+0.051 [-0.070, +0.150]$	0.557 $\rightarrow$ 0.540, $-0.017 [-0.124, +0.070]$	0.806 / 0.804
	test specificity 0.90	0.323 $\rightarrow$ 0.218, $-0.105 [-0.221, +0.019]$	0.343 $\rightarrow$ 0.437, $+0.094 [-0.028, +0.196]$	0.336 $\rightarrow$ 0.364, $+0.028 [-0.070, +0.119]$	0.903 / 0.904
frozen probe	test sensitivity 0.80	0.803 $\rightarrow$ 0.715, $-0.087 [-0.172, +0.018]$	0.800 $\rightarrow$ 0.843, $+0.043 [-0.009, +0.083]$	0.801 $\rightarrow$ 0.801, $+0.000 [-0.005, +0.006]$	0.611 / 0.542
	test specificity 0.80	0.505 $\rightarrow$ 0.331, $-0.174 [-0.313, -0.012]$	0.514 $\rightarrow$ 0.638, $+0.124 [+0.001, +0.231]$	0.511 $\rightarrow$ 0.536, $+0.025 [-0.083, +0.132]$	0.803 / 0.803
	test specificity 0.90	0.231 $\rightarrow$ 0.151, $-0.079 [-0.244, +0.086]$	0.313 $\rightarrow$ 0.390, $+0.077 [-0.040, +0.200]$	0.286 $\rightarrow$ 0.311, $+0.025 [-0.085, +0.141]$	0.903 / 0.904
	test sensitivity 0.80	0.779 $\rightarrow$ 0.713, $-0.067 [-0.155, +0.026]$	0.811 $\rightarrow$ 0.844, $+0.033 [-0.013, +0.075]$	0.801 $\rightarrow$ 0.801, $+0.000 [-0.006, +0.006]$	0.538 / 0.515

Table 19: Descriptive: the other unaltered test sets at matched test specificity 0.80, mask $-$ ERM (crossed 95% CIs); conflicting = positives whose artifact status conflicts with the test set’s own association.

Test set (frozen probe)	Conflicting positives	Aligned positives	All positives
Dermoscopy, BCN held out	$-0.040 [-0.065, -0.015]$	$-0.034 [-0.062, -0.003]$	$-0.038 [-0.057, -0.017]$
Dermoscopy, HAM held out	$+0.067 [+0.032, +0.106]$	$+0.039 [-0.019, +0.095]$	$+0.059 [+0.027, +0.093]$
Dermoscopy, MSK held out	$-0.001 [-0.054, +0.053]$	$+0.193 [+0.067, +0.311]$	$+0.020 [-0.030, +0.070]$
Capsule (held-out frames)	$+0.037 [-0.013, +0.110]$	$+0.144 [+0.107, +0.196]$	$+0.131 [+0.098, +0.179]$

S12. CLAIM 2024 checklist

Checklist for Artificial Intelligence in Medical Imaging, 2024 update [7]. M = main paper section, S = supplement section.

No.	Item	Answer	Where / comment
<i>Title / abstract</i>			
1	Identification as a study of AI methodology, specifying the category of technology used (eg, deep learning)	Yes	Title, abstract: deep-learning classifiers on frozen foundation-model encoders and fine-tuned CNN/ViT.
2	Summary of study design, methods, results, and conclusions	Yes	Abstract.
<i>Introduction</i>			
3	Scientific and/or clinical background, including the intended use and role of the AI approach	Yes	M Introduction, Related work.
4	Study aims, objectives, and hypotheses	Yes	M Introduction; hypotheses pre-registered (S1).
<i>Methods — study design</i>			
5	Prospective or retrospective study	Yes	Retrospective analysis of public datasets (M Data).
6	Study goal	Yes	Methodological study of shortcut mitigation (not a diagnostic-accuracy study); M Introduction, Setup.
<i>Methods — data</i>			
7	Data sources	Yes	M Data (ISIC 2018/2019, HAM10000 masks, TN3K/TNCD, MMOTU, SEE-AI, NIH ChestX-ray14); licences in docs/DATA.md.
8	Inclusion and exclusion criteria	Yes	M Data (trap cells, artifact-free groups, QC); transplant feasibility (S11).
9	Data preprocessing	Yes	M Setup (518 px rendering, ROI masking, feature caching).
10	Selection of data subsets	Yes	Traps, matched traps and transplant cohorts (M Results, S11).
11	De-identification methods	NA	Public datasets released de-identified by their providers.
12	How missing data were handled	Yes	Missing age imputed with an indicator; images without a lesion mask excluded by QC (152 ISIC images).
13	Image acquisition protocol	Partial	As documented by the dataset providers (cited); no acquisition parameters are released for TN3K, MMOTU or SEE-AI.
<i>Methods — reference standard</i>			
14	Definition of method(s) used to obtain reference standard	Yes	Diagnoses as released by the dataset providers (TNCD: cytological biopsy; ISIC: histopathology, follow-up or expert consensus; MMOTU tumour categories; SEE-AI expert lesion annotation) (M Data).
15	Rationale for choosing the reference standard	Yes	Dataset reference standards; ROI masks from manual annotation or a U-Net (M Data).
16	Source of reference standard annotations	Yes	Dataset annotators; artifact labels from a rule-based detector, a patch probe and published masks; full human/automatic provenance in S4.
17	Annotation of test set	Yes	Same procedures as for training images.
18	Measures of inter- and intrarater variability of features described by the annotators	No	Artifact labels were audited by the analysis team only; a blinded two-rater expert audit kit with Cohen's κ is prepared (S4) but not yet carried out (M Limitations).
<i>Methods — data partitions</i>			

No.	Item	Answer	Where / comment
19	How data were assigned to partitions	Yes	Five seeds of five-fold group-safe cross-validation; official TNCD split; leave-one-hospital-out (M Setup, Consequences on unaltered data).
20	Level at which partitions are disjoint	Yes	Lesion identifiers (ISIC), patient level for the official TNCD split and chest radiographs, otherwise perceptual-hash and frame-block groups; embedding-group sensitivity (M Robustness).
<i>Methods — testing data</i>			
21	Intended sample size	No	No a-priori sample-size calculation; all eligible images used; underpowered contrasts are identified (ovary).
<i>Methods — model</i>			
22	Detailed description of model	Yes	Frozen DINOv2 ViT-S/B/L, MedSigLIP, DermLIP, RAD-DINO, ConvNeXt with logistic heads; fine-tuned ResNet-50 and ViT-S (M Setup, Replication; S8).
23	Software libraries, frameworks, and packages	Yes	<code>requirements.txt</code> (PyTorch 2.11, timm, scikit-learn, open_clip) (S13).
24	Initialization of model parameters	Yes	Published pre-trained checkpoints; ImageNet initialisation for fine-tuning.
<i>Methods — training</i>			
25	Details of training approach	Yes	Regularisation and decision threshold on clean validation data; fine-tuning recipe (S8).
26	Method of selecting the final model	Yes	Clean-validation AUROC; no test-set selection.
27	Ensembling techniques	NA	No ensembles.
<i>Methods — evaluation</i>			
28	Metrics of model performance	Yes	Reversed, correlated and clean AUROC; location crossover; AUPRC, Brier, ECE, sensitivity/specificity (M Setup, Consequences on unaltered data).
29	Statistical measures of significance and uncertainty	Yes	Crossed seed×image bootstrap for every interval, from re-generated runs (S15; old-versus-new comparison in the repository); Holm per family (M Setup; S1).
30	Robustness or sensitivity analysis	Yes	Label definitions, leakage grouping, covariate matching, rebuild on new hardware (M Robustness).
31	Methods for explainability or interpretability	Yes	Same-head counterfactual sensitivity to inserting the artifact (M Mechanism).
32	Evaluation on internal data	Yes	All traps and sweeps.
33	Testing on external data	Partial	Dermoscopy: new patients from ISIC 2020 under a pre-registered gate (patient-disjoint folds) and ISIC leave-one-hospital-out; no external cohort for ultrasound or capsule endoscopy (M Limitations; S3).
34	Clinical trial registration	NA	Retrospective; analysis pre-registrations in the repository (S1).
<i>Results</i>			
35	Numbers of patients or examinations included and excluded	Yes	Image counts per cohort, trap cell and transplant feasibility; patients where identifiers exist (M Data; S11).
36	Demographic and clinical characteristics of cases in each partition	Partial	Age, sex, body site and diagnosis for ISIC; lesion size and acquisition covariates for all cohorts (S11); demographics are not released for TN3K, MMOTU or SEE-AI.
37	Performance metrics and measures of statistical uncertainty	Yes	All result tables give 95% CIs.

No.	Item	Answer	Where / comment
38	Estimates of diagnostic performance and their precision	Yes	Sensitivity and specificity at five pre-registered operating points with CIs (M Consequences on unaltered data; S9).
39	Failure analysis of incorrectly classified cases	Yes	Shortcut-conflicting cases; DermLIP without a hair gap; chest radiography boundary case (M Results).
<i>Discussion</i>			
40	Study limitations	Yes	M Discussion and limitations.
41	Implications for practice, including intended use and/or clinical role	Yes	M Discussion; decision guide (M What carries the shortcut decides the fix).
<i>Other information</i>			
42	Provide a reference to the full study protocol or to additional technical details	Yes	Pre-registration documents and registry (S1); report/full/full_report.pdf .
43	Statement about the availability of software, trained model, and/or data	Yes	Code, pre-registrations and result files are public; all datasets are public (M Conclusion).
44	Sources of funding and other support; role of funders	—	To be completed by the authors.

S13. Reproducibility

All numbers in the main paper and this supplement are produced by scripts in the repository (**scripts/**); tables are generated, never typed, and every confidence interval in the paper is traced automatically to a result file (**scripts/verify/audit_paper_numbers.py**). A complete rebuild on different hardware reproduces every primary crossover within 0.01 (main text). Compute: one 24 GB GPU.

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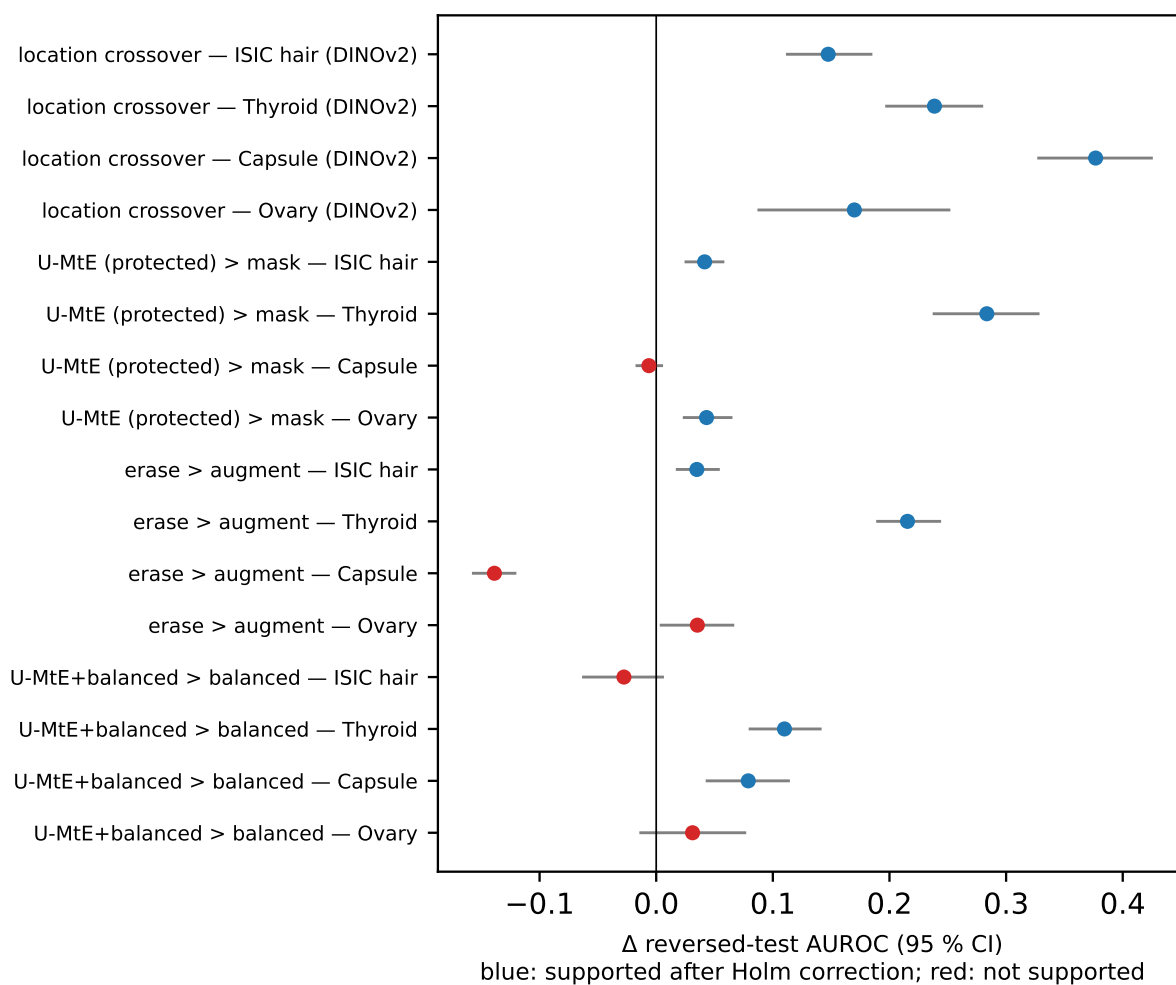


Figure 1: The 16 pre-registered primary tests with Holm correction (blue: supported; red: not supported). Generated by `scripts/analysis/primary_claims.py` and `scripts/make_figures.py`.

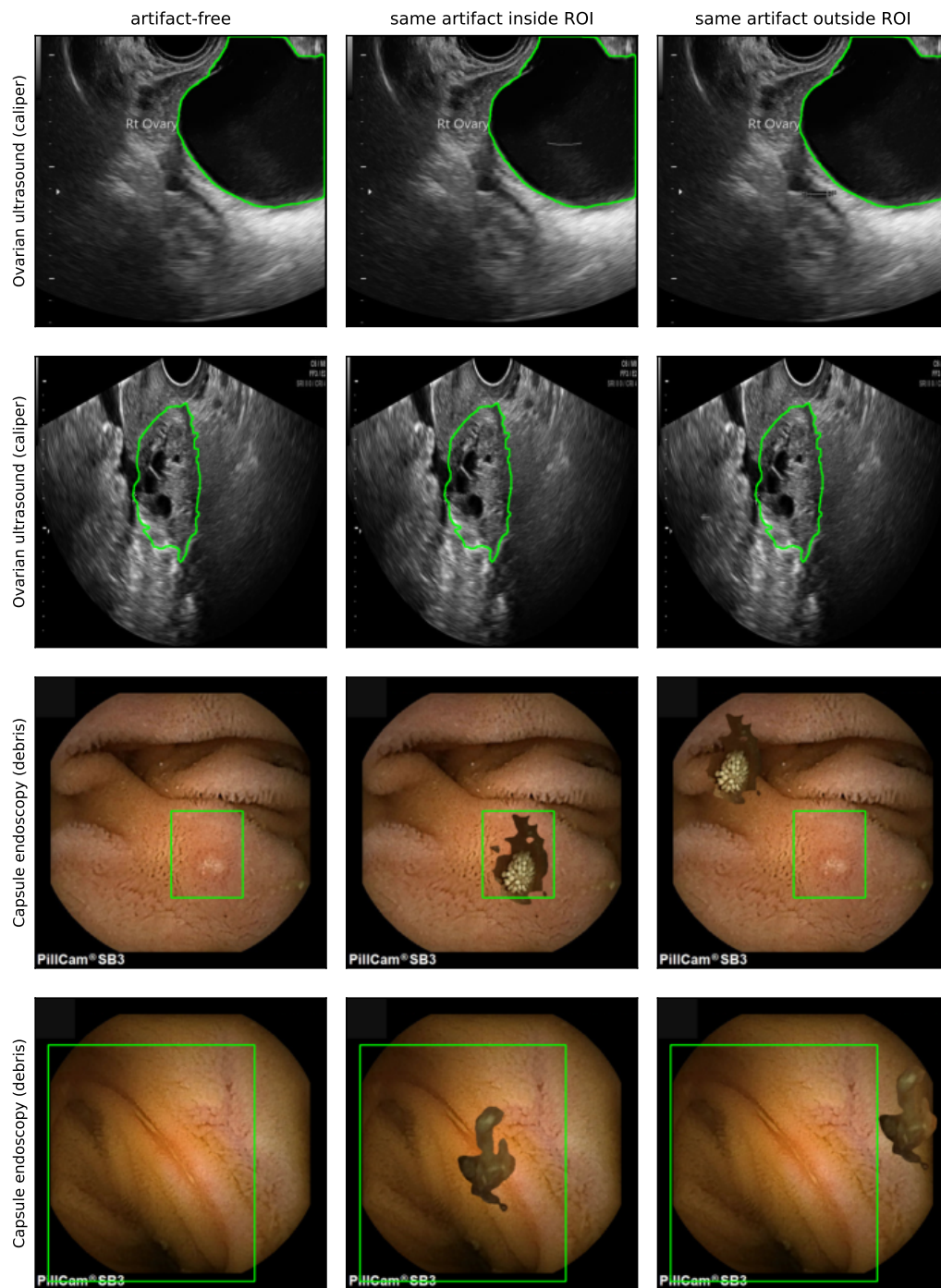


Figure 2: Real-artifact transplant, openly licensed cohorts (green: ROI). Each row shows one artifact-free image and the same real artifact instance—cut from another image of the cohort with its mask—pasted inside and outside the ROI. Instances carry a thin rim of donor background where the mask extends beyond the artifact; the rim is identical at both locations. Generated by `scripts/make_transplant_figure.py`.